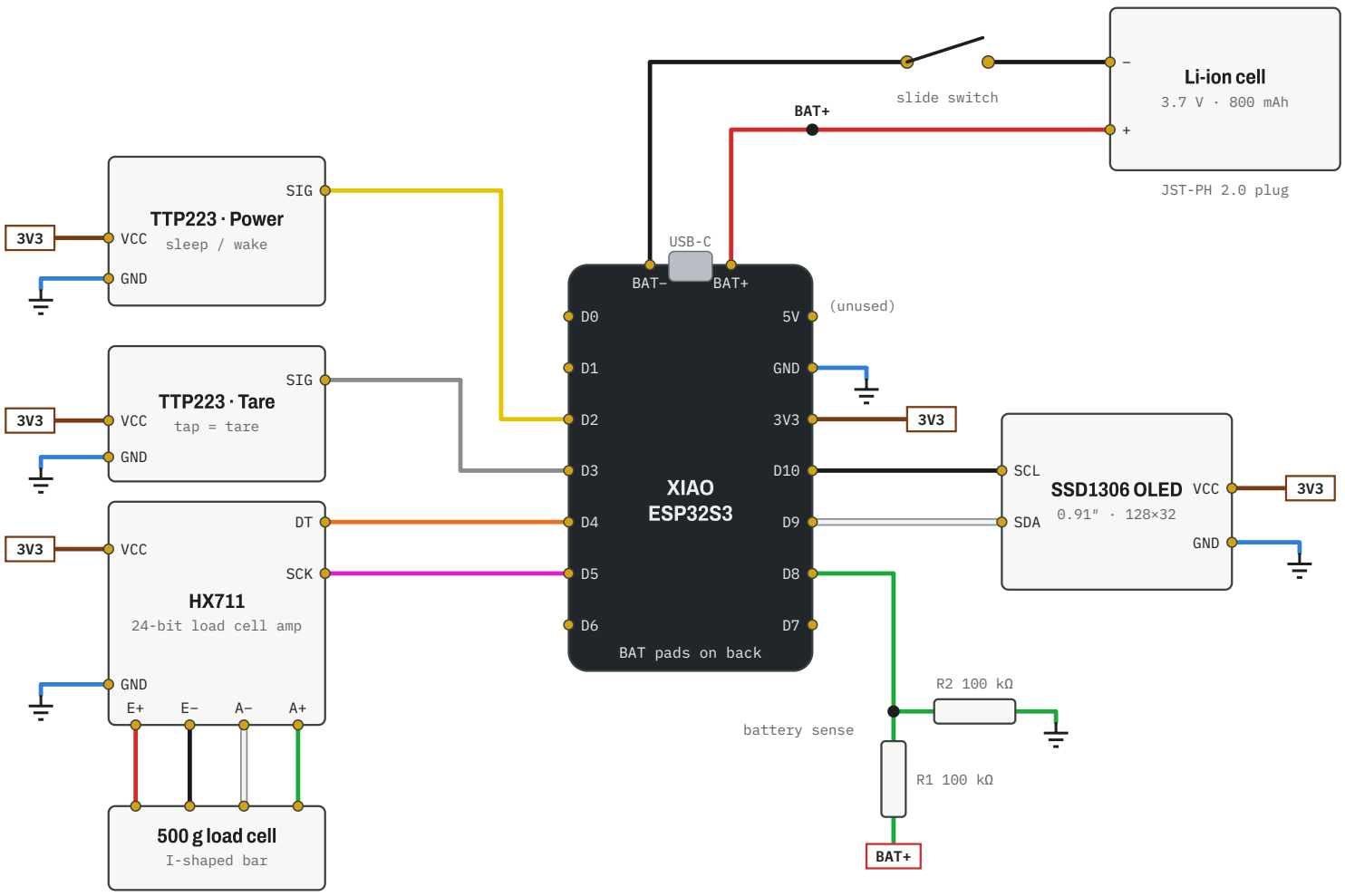
XIAO ESP32S3 BUILD · FIRMWARE V2.2.1

WeighMyBru² XIAO Wiring

Every connection between the Seeed XIAO ESP32S3, HX711, load cell, OLED, two TTP223 touch modules, and the Li-ion battery. GPIO numbers come from `include/BoardConfig.h` and are the same as the SuperMini build. Only the physical pins change. Wire colors match the official XIAO diagram on weighmybru.com. Hover a row in the table to highlight its wire.



● Dot = joined wires. Crossing lines without a dot are not connected. **3V3** flag = XIAO 3V3 pin ⏚ = XIAO GND

Board drawn from the front, USB-C at the top.

Pin map

XIAO PIN	GPIO	GOES TO	WIRE	FIRMWARE NAME
D2	GPIO3	TTP223 Power · SIG	yellow	TOUCH_SLEEP_PIN
D3	GPIO4	TTP223 Tare · SIG	gray	TOUCH_TARE_PIN
D4	GPIO5	HX711 · DT	orange	HX711_DATA_PIN
D5	GPIO6	HX711 · SCK	magenta	HX711_CLOCK_PIN
D8	GPIO7	Divider midpoint (R1/R2)	green	BATTERY_PIN
D9	GPIO8	OLED · SDA	white	I2C_SDA_PIN
D10	GPIO9	OLED · SCL	black	I2C_SCL_PIN
3V3		VCC on HX711, OLED, both TTP223	brown	
GND		GND on HX711, OLED, both TTP223, R2	blue	
BAT+	back	Battery + and R1	red	
BAT−	back	Slide switch, then battery −	black	

Load cell to HX711

LOAD CELL WIRE	HX711 PAD
Red	**E+**
Black	**E−**
White	**A−**
Green	**A+**

Build notes

D4 and D5 go to the HX711, not the OLED

Seeed's pinout labels D4 and D5 as SDA and SCL. This firmware ignores those labels: D4 and D5 drive the HX711, and the OLED uses D9 (SDA) and D10 (SCL). Wiring the OLED to D4/D5 leaves the screen blank.

Battery pads are on the back

Solder the battery leads to the BAT+ and BAT− pads on the underside of the XIAO. The XIAO has a built-in charger, so the battery charges over USB-C. The slide switch sits on the negative lead, the same as the SuperMini build.

Battery sense divider

The XIAO does not route its battery voltage to an ADC pin, so you still need the two 100 kΩ resistors. Their midpoint goes to D8 (GPIO7). The firmware multiplies the reading by 2.0, so a full 4.2 V cell reads 2.1 V at the pin.

Touch pads are TTP223 modules

The firmware reads D2 and D3 with `digitalRead` and an internal pull-down. Each pad needs a TTP223 module with SIG, VCC and GND.

Both touch pins do extra jobs

D2 wakes the scale from deep sleep. Holding the tare pad (D3) during boot clears the saved WiFi credentials.

Unused pins

Leave 5V, D0, D1, D6 and D7 unconnected.

Flash the XIAO firmware

Use the `esp32s3-xiao` build. It sets `BOARD_XIAO` and the 8 MB flash layout.

Sources: `include/BoardConfig.h`, `platformio.ini`, `src/main.cpp`, the official XIAO wiring image in the assembly guide, and Seeed's XIAO ESP32S3 pinout.